

Music Genre Classification using Support Vector Machine Techniques

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Abstract—The classification of musical genres is crucial for enhancing music lovers' listening experiences, considering the vast amount of music available worldwide. This study conducted an in-depth analysis of a dataset obtained from Kaggle, which comprised detailed information on various music genres and their associated features. The dataset was grouped into four genres: pop, rap, rock, and hip-hop. To perform genre classification and compare the accuracy and F1 score, several machine learning algorithms, such as SVM Linear, SVM RBF, SVM Poly, and SVM Sigmoid, were employed. The team identified the most significant features that contributed to the classification process. The results of the study revealed that the SVM Linear algorithm outperformed other SVM kernels in regard to accuracy and F1 score. This finding indicates that SVM Linear is the most effective algorithm for classifying music genres in the dataset under investigation. The results of this study have important ramifications since they provide light on musical genre classification and the potential application of machine learning methods to improve genre identification precision. In addition, this study may have applications for music streaming services, recommendation engines, and other music-related software, providing users with a more personalized and enjoyable music listening experience.

Keywords—classification, music, genre, support vector machine, SVM, accuracy, F1 score, computer science

I. INTRODUCTION

With thousands of music in this modern world, it is important to classify them into groups, also known as genres. This is important because different musical genres give singers and their music a unique personality. By doing this, it enables listeners to quickly categorize and distinguish between the types of music that they enjoy the most compared to those that do not fit them. Classifying music genres can also help artists to target a specific audience, which can result in increased popularity and success on their part [1].

This study explores the application of machine learning algorithms in solving the given problem. It is common knowledge that classification, a supervised learning concept in machine learning, separates a set of data into classes [2]. In other words, machine learning can classify a collection of music into classes or genres. The ability to categorize a music's genre enables many kinds

of music to be compiled or recommended based on its genre. This would be beneficial for audio streaming services, such as *Spotify* or *Joxx*, and also for music charts like *Billboards*. A common method used to classify a music's genre is by using metrics obtained from *Spotify*'s organized playlist. Moreover, this study aims to explore one of the implementations of a machine learning algorithm, which is Support Vector Machine (SVM). SVM is a supervised classification algorithm to differentiate between categories [3]. It excels when the data is unknown and works well with unstructured and semi-structured data[4].

This situation can highlight several advantages of the 4 different types of SVM algorithms, namely linear, RBF, sigmoid, and polynomial kernels. Furthermore, the team also investigates the correlative features between one and another with the given dataset.

The remainder of this paper is structured as follows. The next section will go through how this research differs from other works of existing literature that use comparable methodologies. Next, there will be an overview of the dataset and its preprocessing to suit the research. The algorithms and evaluation method used will be the next discussion. Finally, the result and discussion will be the last part of this study, followed by the conclusion and the recommendation for future research.

II. LITERATURE REVIEW

Music genre classification has been in numerous studies run by many people from various times. In April 2018, Kehan Luo [5] published a paper titled "Machine Learning Approach for Genre Prediction on Spotify Top Ranking Songs". Between January 2017 and August 2017, this article examined the audio attributes and genres of the top-performing songs on Spotify. The paper's analysis includes summary statistics, principal component analysis, and machine learning classifier installation and evaluation. Kehan Luo investigated the major component by dividing nine audio features into three. The machine learning technique used genres and audio qualities as input and provided genre predictions based on audio features for the

test set of songs. Finally, the classifier obtained a 46.9% accuracy rate.

There exists a project report named “Music Genre Classification”, created by Robert and Yuan Hong. This study discovered that because the differences between musical genres are largely arbitrary, genre classification has not yet been automated by conventional algorithmic approaches. Robert and Yuan thought that with enough audio data, which can be easily extracted from large amounts of freely available music online, machine learning can observe and make predictions using these ill-defined patterns [6]. From that, the study explains the project’s objective in predicting the genre and confidence level of classical, jazz, rap, and rock songs through 1D CNN, 2D CNN, and multi-layer GRU.

Finally, Chaudary, Aziz, Khan, and Gretschmann published a study in 2021 on the classification of music genres using the Support Vector Machine approach. The research paper is named ”Music Genre Classification using Support Vector Machine and Empirical Mode Decomposition”. To extract the audio signal in this work, the first eight Intrinsic Mode Functions (IMFs) and Empirical Mode Decomposition (EMD) were used [7]. To obtain the best accuracy, the Support Vector Machine (SVM-L) is then employed and trained utilizing data from five diverse genres. Summaries can be seen in TABLE I

TABLE I. LITERATURE DISCUSSION

Literature paper List		
Author	Title	Algorithm
Kehan Luo	Machine Learning Approach for Genre Prediction on Spotify Top Ranking Songs	One Vs Rest Classifier
Robert and Yuan Hong	Music Genre Classification	1D CNN, 2D CNN, and multilayer GRU
Chaudary et al.	Music Genre Classification using Support Vector Machine and Empirical Mode Decomposition	Linear SVM

III. METHODOLOGY

According to the previous section, several of the works have a similar goal to this study. The key difference is that the goal of this study is to present a comparison perspective between 4 types of SVM utilizing a Kaggle dataset. For comparison, none of the research use the same dataset or techniques. Except for Linear SVM, none of the other algorithms used in previous publications were used in this study. Another point to notice is that this research has the least quantity of data when compared to the cited studies because the dataset comes from a single source, which is a single Kaggle dataset. Finally, only two features are used in this study. According to our research, no other work has specifically experimented with the same algorithms and datasets used in this study.

In Fig 1, it shows all of our steps in doing this research. In addition, it also displays the tools and methods the team use to finish and deliver the best results.

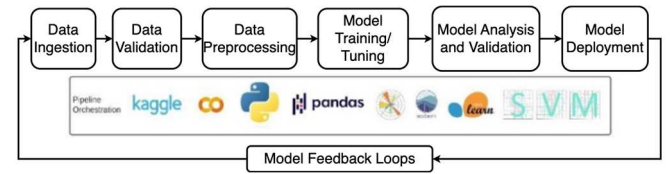


Fig. 1. Pipeline

A. Data Collection

The dataset used is titled “Top 100 Most Streamed Songs on Spotify” created by Pavankalyan, from the Kaggle website [8]. The author extracted 100 songs using *organizeyourmusic* website and split them into 14 explorable features. The dataset features will be explained in TABLE II and for the visualization of the features, the team shows several Histogram graph which can be seen in Fig. 2.

TABLE II. FEATURES DEFINITION

No.	Features	Definition
1	Index	The order of the song’s data.
2	Title	The Name of the song.
3	Artist	The main artist(s) were involved in making the song.
4	Top genre	All top genres that are often played on Spotify.
5	Year	The initial year the song was released.
6	Beats per minute	The tempo of each song.
7	Energy	A perceptual measure of intensity and activity. Energetic tracks feel, fast, loud, and noisy.
8	Danceability	Describes how appropriate a song is for dancing based on a combination of musical elements including tempo, rhythm stability, beat strength, and overall regularity.
9	Loudness	The overall loudness of a song in decibels (dB). The average level of loudness for the entire song ranges from -60 to 0 dB.
10	Liveness	Detects the presence of an audience in the recording. Higher values indicate a higher likelihood that the song was performed live.
11	Valance	Calculates the mood level generated from a song. Higher values represent a more positive mood for the song.
12	Length	The duration of a song.
13	Acousticness	Describes how acoustic a song is. Higher values represent more acoustic the song is.
14	Speechiness	Detects the presence of spoken words in a song. Higher values represent more spoken words the song contains.
15	Popularity	Measure the popularity of the song. Higher values represent the more popular the song is.

B. Data processing

Before modifying the dataset in any way, it is described and displayed to give the team a better understanding of the features and specifications it contains. The team also checked for missing values, which were not found in this dataset. One column was renamed for a better naming convention. For the prediction target label, a new column named “genre class” was added. The values inserted in this column range from 0 to 3 based on the genre identified from the ‘top genre’ column. Only genres that were considered to be either pop, rap, rock, or hip-hop were

included. Genres that were not considered compatible were removed from the data frame, leaving the dataset with only 88 songs. TABLE III shows the detail of compatible genre count.

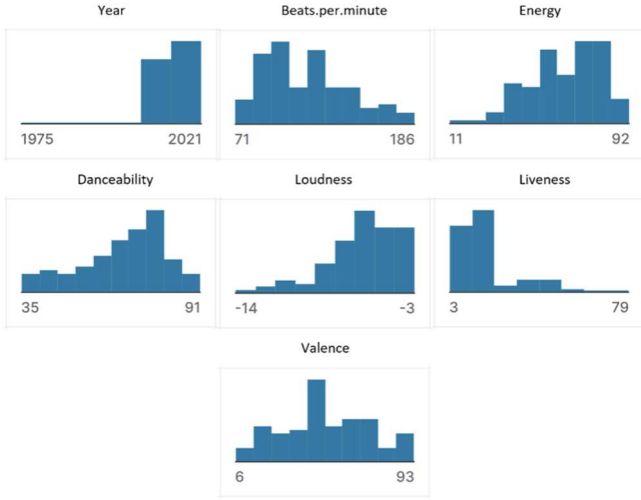


Fig. 2. Histogram Visualization of Features

TABLE III. TOTAL COMPATIBLE GENRE COUNT

Genre	Count
Pop (0)	57
Rap (1)	18
Rock (2)	7
Hip Hop (3)	6

C. Algorithm and Techniques

For the algorithm and techniques that will be used for the dataset, the team chose Support Vector Machines with 4 different types of kernels to search for the best algorithm that can be implemented in solving the problem. The reason for choosing SVM instead of other algorithms is that SVM is a supervised classification algorithm that can distinguish between categories [3]. SVM can classify categories, which in this particular research is the genre of music.

In order to choose the best features for the input values, a feature selection technique is used to show the correlation between the data sets. "The purpose of the feature selection technique is to locate the best set of features that will allow one to design helpful algorithms for the examined phenomena" [9]. A formula called Pearson's correlation formula is used to visualize the correlation, which is shown below

$$\frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum(x_i - \bar{x})^2 \sum(y_i - \bar{y})^2}} \quad (1)$$

The value of the correlation coefficient can be any value in the range of -1 and 1

- It is believed that if the value is 1, there is a positive correlation between two variables. This means that when one variable increases in value, so does the

other.

- A negative correlation between the two variables is said to exist if the value is -1. This means that as one variable rises in value, the other decreases.
- There is no relationship between the two variables if the value is 0. This means that the variables' associations to one another change at random.

A variety of tools were used to accomplish this research, which are listed below:

- *Pandas*: read and modify the dataset
- *Matplotlib*: set or control the visualization of the dataset.
- *Seaborn*: visualize the correlation heatmap between features.
- *Sklearn*: algorithm and data prediction algorithm.

The data set for this study is divided into a training set and a test set, each with a ratio of 7:3. However, before applying it to the machine learning algorithms, *StandardScaler* was applied to the training set. The reason is that scaling variables is necessary before executing the final SVM algorithm because all SVM kernel approaches depend on distance [10].

There was one tool that the team tried using, but didn't use it at the end, which was *mpl_toolkit*. The library allows users to plot data in three dimensions. However, the usage doesn't seem necessary and could be burdensome.

D. Evaluation Method

Evaluating the algorithm is something necessary to do. Through evaluating the algorithm, the team can reduce and eliminate doubts about the applicability of the algorithm to the problem. Finding the accuracy of the algorithm and demonstrating that the machine learning method is the most appropriate for the case are two additional purposes of an evaluation. There are various ways to evaluate an algorithm, but the following metrics are the ones used in this study.

- **F1 score**: An evaluation metric that measures algorithm accuracy by combining the precision and recall score of an algorithm.
- **Accuracy**: A percentage of a correct prediction made by an algorithm and used to indicate how accurate the algorithm predicts the data.

IV. RESULTS AND DISCUSSIONS

A. Result of feature selection

Based on the purpose of using feature selection, Fig. 3 and 4 explained about correlation and direction signed of the modified dataset.

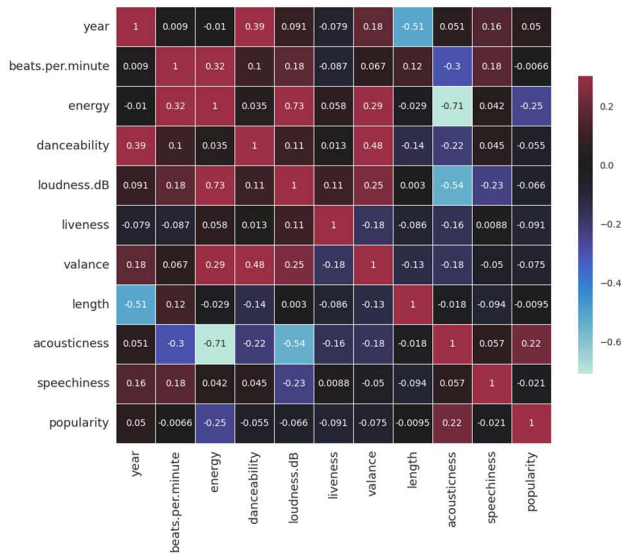


Fig. 3. Seaborn heatmap visualization of the modified dataset

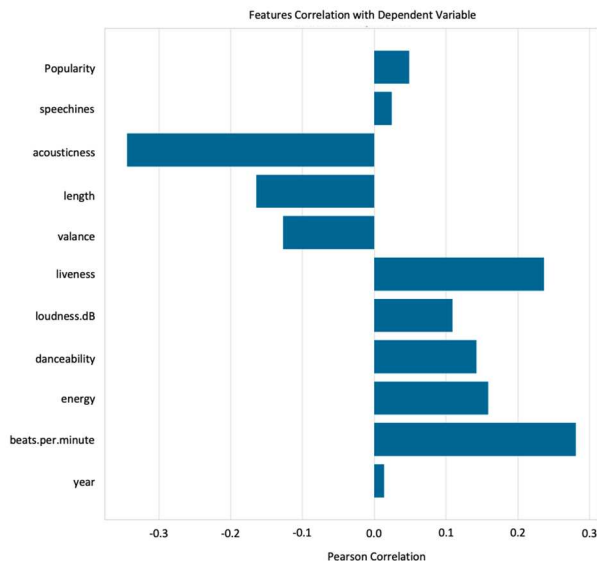


Fig. 4. Histogram Visualization of The Modified Dataset

To know whether the correlation is high or low, see the parameter next to the plot image in Fig 3. This parameter shows the colors that aim to show the correlation between features. Parameters start with the number 0.2 in red and end with the number -0.6 in blue. Brighter color in the plot means a higher correlation number displayed, or above 0.2 and -0.6. In this heatmap, there are the top 3 highest correlation features. First, *beats.per.minute* shows a correlation number of 0.28 with a lighter red color. Second, *liveness* has a correlation number of 0.24 with a slightly darker red color than *beats.per.minute*. Last but not least, *acousticness* has the highest correlation with a value of -0.34.

However, even though *acousticness* has the highest correlation value, this value has a negative value, so only two of the top three highest correlation features were chosen as the inputs, which of course have a positive value.

B. Evaluation result

Total of 88 samples were used in the analysis of the data set, with 70% serving as the training set and the remaining

30% serving as the test set. Several C and gamma parameters were evaluated in order to discover the appropriate parameter utilization.

The input values of the C and Gamma parameters in all SVM kernel algorithms are 1. It can also be inferred that lower C and gamma values result in higher F1 scores. In terms of F1 score, the Linear and RBF kernels perform better than the Sigmoid and Poly kernels with a difference of 0.06. However, it is clear that overall, the SVM algorithms delivers an acceptable accuracy for this dataset.

Speaking of accuracy, none of the kernels was able to predict with a minimum accuracy of 80%. However, it is seen that the Linear kernel is able to provide the best accuracy when compared to the other kernels used in this research.

TABLE IV. COMPARISON OF EVALUATION FOR EACH SVM KERNELS'

Kernels	F1 Score	Accuracy
Linear	0.88	0.78
RBF	0.88	0.70
Sigmoid	0.82	0.74
Poly	0.82	0.67

From TABLE IV, the F1 scores are better in evaluating the algorithm than the accuracy. "Accuracy is the absence of bias in statistical terms. In other words, measurements are not systematically exaggerated or exaggerated. However, accuracy tells us nothing about the target's distance. Accuracy does not consider how close a measurement is to the target. Instead, it assesses how accurate they are on average" [11]. More importantly, the F1 score is recommended to be used when the data set is unbalanced and is the standard for reliability [12]. This is indeed the case with the dataset chosen for this project, where the class distribution is highly unbalanced.

C. Data distribution from each SVM kernel

From the results of the analysis of different kernel types and parameters in the SVM, the team can determine which kernel type is best at predicting the music genre for this particular data set. The visualization of the best type for each kernel can be seen in the images below.

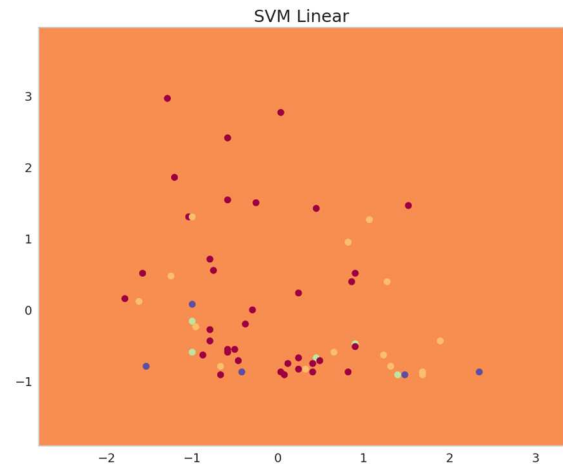


Fig. 5. Data Distribution Visualization of Linear Kernel

- **SVM Linear:** Fig. 5 shows the result based on the first algorithm, the SVM Linear. For the parameter, the team has done some experiments and found that the best parameter is between c and $\gamma = 1$.

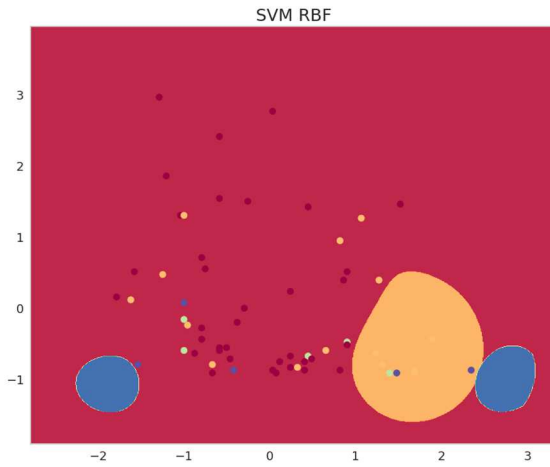


Fig. 6. Data Distribution Visualization of RBF Kernel

- **SVM RBF:** Fig. 6 shows the second algorithm used in this study, which is SVM RBF. For the parameter, the team has done some experiments and found that the best parameter is between c and $\gamma = 1$.

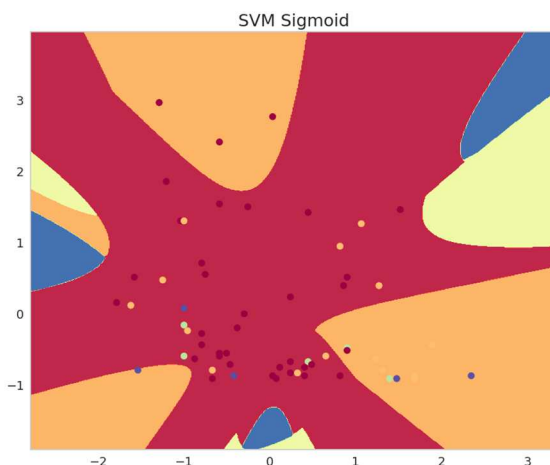


Fig. 7. Data Distribution Visualization of Sigmoid Kernel

- **SVM Sigmoid:** Fig. 7 shows the third algorithm, SVM with sigmoid kernel. For the parameter, the team has done some experiments and found that the best parameter is between c and $\gamma = 1$.

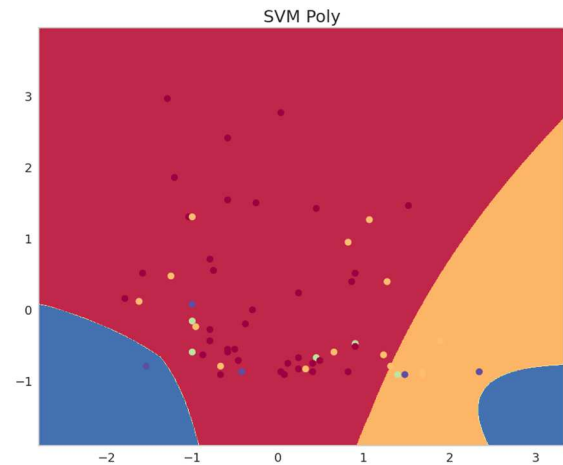


Fig. 8. Data Distribution Visualization of Polynomial Kernel

- **SVM Polynomial:** Fig. 8 shows the last algorithm that was used in this project, which is the SVM with a polynomial kernel. For the parameter, the team used the grid search method to look for the best parameter by experimenting with several points, C : [0.1, 1, 100, 1000] and degree: [1, 2, 3, 4, 5, 6].

D. Result of algorithm prediction

After evaluating the algorithm based on the results, the team can determine how good the algorithm is performing. To demonstrate how the algorithm prediction works, refer to the provided examples below.

TABLE V. RESULT OF ALGORITHM PREDICTION

Prediction Details				
SVM	Test 1	Result	Test 2	Result
Linear	BPM=100, Liveness=9, Acousticness=7	Correct	BPM=114, Liveness=17, Acousticness=25	Correct
RBF	BPM=100, Liveness=9, Acousticness=7	Correct	BPM=114, Liveness=17, Acousticness=25	Correct
Sigmoid	BPM=100, Liveness=9, Acousticness=7	False	BPM=114, Liveness=17, Acousticness=25	False
Poly	BPM=100, Liveness=9, Acousticness=7	Correct	BPM=114, Liveness=17, Acousticness=25	Correct

In predicting the class genre, the team filled the dummy data with selected features based on the feature selection method. Only *beats.per.minute*, *liveness*, and *acousticness* were selected for use because they have a high correlation with one another. From the prediction results in TABLE V, it is known that in the first and second predictions, both Linear SVM, Poly SVM, and RBF SVM can predict the correct genre class, namely pop. Only Sigmoid SVM has a different or possibly incorrect prediction, with the predicted result of the class rock genre.

V. CONCLUSION

This study examines the role of music genre classifications using the given dataset. The ability to predict a song's genre is useful in organizing songs to assist music artists. We proposed a solution by using Support Vector

Machine to classify the music genre. There are four different types of SVM used, which are Linear, RBF, Sigmoid, and Polynomial. The first step in this study was to process the data set and remove unnecessary or incompatible data. The second step involved employing the feature selection technique and evaluating the algorithm to choose the best features. The *beats.per.minute*, *liveness*, and *acousticness* were determined to be the most important features in the data set. After the training and tuning process, the algorithm was used to predict the music genre. Although the machine learning algorithm isn't perfect to predict a song's genre, it is still possible to predict a song's genre with good accuracy. In this study, the Linear SVM algorithm outperformed other SVM algorithms. It can be seen in the results.

The dataset used in this study was obtained from Kaggle. The dataset is small when compared to other datasets. For future research, something to take note of is to expand the findings would be to increase the size of a dataset and make comparisons between multiple datasets. That should give a better insight into how the size of a dataset could affect the prediction of the algorithm.

REFERENCES

- [1] J.J. Aucouturier, F. Pachet F. "Representing musical genre: A state of the art". Journal of new music research. 2003 Mar 1;32(1):83-93.
- [2] N. Ndou, R. Ajoodha, A. Jadhav. "Music genre classification: A review of deep-learning and traditional machine-learning approaches". In 2021 IEEE International IOT, Electronics and Mechatronics Conference (IEMTRONICS) 2021 Apr 21 (pp. 1-6).
- [3] M. Awad and R. Khanna. "Support vector machines for classification". Efficient Learning Machines: Theories, Concepts, and Applications for Engineers and System Designers. 2015:39-66.
- [4] D. Anguita, A. Ghio, N. Greco, L. Oneto, S. Ridella. "Model selection for support vector machines: Advantages and disadvantages of the machine learning theory". In The 2010 international joint conference on neural networks (IJCNN) 2010 Jul 18 (pp. 1-8).
- [5] K. Luo, "Machine Learning Approach for Genre Prediction on Spotify Top Ranking Songs," cdr.lib.unc.edu, Feb. 28, 2019. <https://cdr.lib.unc.edu/concern/masters-papers/ns064961b>
- [6] R. Adragna, & Y. Hong (Bill) Sun. "Music Genre Classification". (n.d.). <https://www.eecg.utoronto.ca/~jayar/mic324/musicgenre.pdf> (accessed: Jan. 16, 2023).
- [7] E. Chaudary, S. Aziz, M. U. Khan and P. Gretschnann. "Music Genre Classification using Support Vector Machine and Empirical Mode Decomposition," 2021 Mohammad Ali Jinnah University International Conference on Computing (MAJICC), Karachi, Pakistan, 2021, pp. 1-5, doi: 10.1109/MAJICC53071.2021.9526251.
- [8] Pavankalyan, "Top 100 Most Streamed Songs on Spotify", Kaggle, 2021. <https://www.kaggle.com/datasets/pavan9065/top-100-most-streamed-songs-on-spotify> (accessed Jan. 16, 2023).
- [9] S. Ray. "A quick review of machine learning algorithms". In 2019 International conference on machine learning, big data, cloud and parallel computing (COMITCon) 2019 Feb 14 (pp. 35-39).
- [10] S. Vinay. "Standardization in machine learning." LinkedIn. Jan. 26, 2021. <https://www.linkedin.com/pulse/standardization-machine-learning-sachin-vinay/?trk=public>
- [11] C.V. Deutsch, "Direct assessment of local accuracy and precision." Geostatistics wollongong 96.1 (1997): 115-125.
- [12] R. Yacoub, and D. Axman, November. Probabilistic extension of precision, recall, and f1 score for more thorough evaluation of

classification models. In Proceedings of the first workshop on evaluation and comparison of NLP systems, 2020 (pp. 79-91).